

SUMMER SCHOOL ON DIFFERENTIAL GEOMETRY IN XIAMEN

# Harmonic Map Theory——Note

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Based on *Topics in Analysis: Harmonic Map Theory*,

Xin Zhou, June 25, 2026.

I just write down what has been mentioned on the summer school so the logic may not be coherent. Futhermore, the proof merely states the train of thought

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# 1 Energy and the definition of a harmonic map

**Source:** §1.1, pp. 3–4; Definitions 1.1–1.3 and equation (1.1).

Let  $(M^m, g)$  and  $(N^n, h)$  be smooth Riemannian manifolds, with  $N$  closed as in the course's basic setup. Interior variations are compactly supported; when a boundary is present, boundary values are fixed.

**Definitions 1.1 and 1.2: Dirichlet energy density and energy.**

For  $u \in C^1(M, N)$ ,

$$e(u) = \frac{1}{2} |du|_{\text{HS}}^2 = \frac{1}{2} \text{tr}_g(u^*h),$$

$$E(u) = \int_M e(u) \, dV_g.$$

In coordinates  $x^\alpha$  on  $M$  and  $y^i$  on  $N$ ,

$$(u^*h)_{\alpha\beta} = h_{ij}(u) \partial_\alpha u^i \partial_\beta u^j, \quad e(u) = \frac{1}{2} g^{\alpha\beta} h_{ij}(u) \partial_\alpha u^i \partial_\beta u^j.$$

Repeated indices are summed. Intrinsically,

$$du \in \Gamma(T^*M \otimes u^*TN), \quad (u^*TN)_x = T_{u(x)}N.$$

The differential is a linear map between two different tangent spaces. The Hilbert–Schmidt norm adds the squared lengths of  $du(e_a)$  over an orthonormal basis of the domain. Energy measures total first-order stretching; it does not measure the size of the image alone. The pullback bundle lets us differentiate vector fields attached to the moving point  $u(x)$ .

**Definition 1.3: Harmonic map.**

A map is harmonic if it is a critical point of  $E$ :

$$\left. \frac{d}{dt} \right|_{t=0} E(u_t) = 0$$

for every admissible variation  $u_t$  with  $u_0 = u$ . Equivalently, one may test  $u_t(x) = \exp_{u(x)}(tV(x))$  with  $V \in \Gamma_c(u^*TN)$ . A critical point need not minimize energy. The condition says that every allowed infinitesimal change has zero first-order energy change. Curvature of the target enters because  $V(x)$  must remain tangent to  $N$ .

The source first defines  $e = \frac{1}{2} |du|^2$ , but drops 1/2 in equation (1.1) and later passages. These notes keep the definition above. To compare with the later unhalved convention, write  $\mathcal{E}(u) = \int |du|^2 = 2E(u)$ .

## 2 The Euler–Lagrange equation: three equivalent forms

**Source:** §1.2, pp. 4–9; Propositions 1.4–1.7, equations (1.2)–(1.6).

### 2.1 Intrinsic first variation and the tension field

**Proposition 1.4.**

The weak first-variation equation is

$$\int_M \langle du, \nabla^u V \rangle dV_g = 0 \quad \text{for all } V \in \Gamma_c^1(u^*TN).$$

For  $u \in C^2$ , it is equivalent to

$$\tau(u) := \operatorname{tr}_g \nabla^{M,u} du = 0.$$

Here

$$(\nabla_X^{M,u} du)(Y) = \nabla_X^u(du(Y)) - du(\nabla_X^M Y).$$

**Proof idea.** Put  $F(t, x) = u_t(x)$  and  $V = \partial_t F|_0$ . Metric compatibility and the torsion-free identity give  $D_t du_t = \nabla^{u_t} \partial_t u_t$ . Hence

$$\left. \frac{d}{dt} \right|_0 E(u_t) = \int_M \langle du, \nabla^u V \rangle = - \int_M \langle \tau(u), V \rangle.$$

The boundary term vanishes because  $V$  is compactly supported or has zero boundary values. Arbitrary  $V$  forces the tension field to vanish.

The tension field is the trace of a covariant Hessian. For a real function it is the Laplacian; for a curve it is covariant acceleration. Thus harmonic functions and affinely parametrized geodesics are the first examples of harmonic maps.

**2.2 Coordinate equation and regularity****Proposition 1.5.**

In target coordinates, for  $i = 1, \dots, n$ ,

$$\Delta_g u^i + g^{\alpha\beta} \Gamma_{jk}^i(u) \partial_\alpha u^j \partial_\beta u^k = 0, \quad \Delta_g f = \frac{1}{\sqrt{\det g}} \partial_\alpha (\sqrt{\det g} g^{\alpha\beta} \partial_\beta f).$$

**Proof idea.** Differentiate the coordinate energy under  $u_t^i = u^i + t\varphi^i$ . Integrate the terms containing  $\partial\varphi$  by parts. The remaining derivatives of  $h$  combine into

$$h_{\ell i} \Gamma_{jk}^\ell = \frac{1}{2} (\partial_j h_{ik} + \partial_k h_{ij} - \partial_i h_{jk}).$$

Invert  $h$  to isolate each equation.

The principal part is the scalar Laplace–Beltrami operator in each component. Nonlinearity is quadratic in first derivatives; this is why mere  $W^{1,2}$  control is delicate.

**Proposition 1.6.**

A  $C^1$  harmonic map between smooth Riemannian manifolds is smooth in the interior.

**Proof idea.** The coordinate right-hand side is locally bounded. Interior elliptic estimates give  $W^{2,p}$  for every finite  $p$ . Choose  $p > m$  to obtain  $C^{1,\alpha}$ , then use Schauder estimates and bootstrap. This argument starts with  $C^1$  regularity, not just a weak  $W^{1,2}$  solution.

**2.3 Extrinsic equation and a sign convention**

Isometrically embed  $N \subset \mathbb{R}^L$ . Write  $D$  for the Euclidean connection and adopt the source's convention

$$A(X, Y) = -(D_X Y)^\perp, \quad X, Y \in TN.$$

**Proposition 1.7.**

For a  $C^2$  map into  $N$ ,

$$u \text{ harmonic} \iff \Delta_g u \perp T_{u(x)}N \iff \boxed{\Delta_g u + \operatorname{tr}_g A(u)(du, du) = 0.}$$

**Proof idea.** If  $P(u)$  is orthogonal projection onto  $T_u N$ , then  $\tau(u) = P(u)\Delta_g u$ . The normal part satisfies  $(I - P(u))\Delta_g u = -\operatorname{tr}_g A(u)(du, du)$  by differentiating  $\Pi_N \circ u = u$  twice. These two projections give the result.

A constrained critical point has no tangential Laplacian. Its normal Laplacian need not vanish: it is exactly the correction needed to keep the map on the curved target. For the unit sphere,

$$A_u(X, Y) = \langle X, Y \rangle u, \quad \Delta_g u + |du|^2 u = 0.$$

**Editorial note.** Orthogonal projection is  $d\Pi_N$  at points of  $N$ ; it is not generally an orthogonal projection at every point of a tubular neighborhood, as the wording on p. 7 might suggest. Taking the inner product of the sphere equation with  $u$  agrees with  $\Delta|u|^2 = 0$ :  $\langle \Delta u, u \rangle = -|du|^2$ .

### 3 Bochner identity, second variation, and NPC convexity

**Source:** §1.3 and §1.3.1, pp. 9–11; Proposition 1.8, equations (1.7)–(1.9).

For clarity, use the curvature convention

$$R(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z, \quad \operatorname{Rm}(X, Y, Z, W) = \langle R(X, Y)W, Z \rangle.$$

Then  $\operatorname{Rm}(X, Y, X, Y)$  is positive on a positively curved orthonormal plane. Let  $e_a$  be a local orthonormal frame on  $M$  and  $u_a = du(e_a)$ .

**Proposition 1.8: Bochner formula.**

If  $u$  is harmonic,

$$\boxed{\Delta_g e(u) = |\nabla du|^2 + \sum_a \langle du(\operatorname{Ric}^\sharp e_a), u_a \rangle - \sum_{a,b} \operatorname{Rm}^N(u_a, u_b, u_a, u_b).}$$

**Proof idea.** Differentiate  $\frac{1}{2}|du|^2$  twice. One term is  $|\nabla du|^2$ . Commuting the remaining covariant derivatives produces source Ricci curvature and target curvature. The term  $\langle \nabla \tau(u), du \rangle$  vanishes for a harmonic map.

Curvature controls energy density. Source Ricci curvature and target sectional curvature enter with opposite signs. If  $\operatorname{Ric}^M \geq 0$  and  $K^N \leq 0$ , then  $e(u)$  is subharmonic. On a closed domain, integration forces the nonnegative terms to vanish. On a general compact geometry, the less precise but useful inequality is

$$\Delta|du|^2 \geq -C(|du|^2 + |du|^4).$$

**Second variation at a harmonic map.**

For an admissible variation with velocity  $V$ ,

$$\boxed{\delta^2 E_u(V, V) = \int_M \left( |\nabla^u V|^2 - \sum_a \operatorname{Rm}^N(V, u_a, V, u_a) \right) dV_g.}$$

**Proof idea.** Differentiate the first variation again and commute the  $t$  and  $M$  derivatives. Before assuming harmonicity, an acceleration term  $-\int \langle \tau(u), D_t \partial_t u_t \rangle$  remains. It disappears at a harmonic map, leaving a quadratic form depending only on  $V$ .

Positive target curvature can supply negative directions of the Hessian. Nonpositive curvature makes this quadratic form nonnegative. The source's  $\frac{1}{2}E''$  in (1.8) corresponds to the unhalved energy  $\mathcal{E}$ ; with our normalization the formula above is  $E''$  itself.

### Section 1.3.1: Convexity in nonpositive curvature.

Suppose  $K^N \leq 0$  and  $F(t, x)$  is a *pointwise geodesic* homotopy. Then

$$\frac{d^2}{dt^2} E(F_t) \geq 0.$$

If  $F_0 = u$  is harmonic and the homotopy is admissible, then  $E(F_1) \geq E(u)$ .

**Proof idea.** Along each target geodesic  $D_t \partial_t F = 0$ , so the acceleration term vanishes at every  $t$ , even when  $F_t$  is not harmonic. The remaining terms have nonnegative sign. A convex function with derivative zero at  $t = 0$  cannot decrease for  $t \geq 0$ . Convexity does not hold along arbitrary homotopies. A complete, simply connected NPC target admits unique geodesics between any two points. For a nonsimply connected target, an appropriate lift and homotopy class must be specified; pointwise minimizing geodesics cannot simply be assumed to form a continuous homotopy.

## 4 Weak harmonic maps and stationary harmonic maps

**Source:** §1.4–1.5, pp. 11–12; Definitions 1.9–1.11.

### Sobolev maps into a target manifold.

For  $N \subset \mathbb{R}^L$ ,

$$W^{1,p}(M, N) = \{u \in W^{1,p}(M, \mathbb{R}^L) : u(x) \in N \text{ a.e.}\}.$$

When  $p > m = \dim M$ , Sobolev embedding gives continuous representatives; locally the Hölder exponent is  $1 - m/p$ . When  $p \leq m$ , continuity does not follow from the Sobolev norm.

### Definition 1.10: Weak harmonicity (sign reconciled).

A map  $u \in W^{1,2}(M, N)$  is weakly harmonic if

$$\int_M \langle du, d\eta \rangle dV_g - \int_M \langle \text{tr}_g A(u)(du, du), \eta \rangle dV_g = 0$$

for bounded  $W_0^{1,2}$  ambient test fields  $\eta$  supported in the interior. Equivalently, the extrinsic harmonic map equation holds in distributions. With  $A = -(D_X Y)^\perp$  and  $\Delta = \text{div } \nabla$ , the second term here has a *minus* sign. Definition 1.10 in the source prints a plus sign, which conflicts with equation (1.4). Replacing  $A$  by  $-A$  would reverse both conventions consistently.

### Definitions 1.9 and 1.11: Domain stationarity and minimization.

Domain stationarity means

$$\left. \frac{d}{ds} \right|_{s=0} E(u \circ \phi_s) = 0 \quad \text{for all } X \in \Gamma_c(TM),$$

where  $\phi_s$  is the flow of  $X$ . An energy-minimizing map satisfies

$$\int_\Omega |du|^2 \leq \int_\Omega |dv|^2$$

for every relatively compact comparison domain and every  $N$ -valued competitor with  $v - u \in W_0^{1,2}(\Omega, \mathbb{R}^L)$ .

Target variations change values of the map; domain variations move where those values occur. Minimizers admit both types of variation. For smooth harmonic maps, domain stationarity follows from harmonicity. For weak maps,  $du(X)$  has only  $L^2$  regularity and cannot automatically be used as a target variation.

$$\text{local minimizer} \implies \text{weakly harmonic and domain-stationary.}$$

Here “stationary harmonic” means *both* weakly harmonic and domain-stationary. The source’s Definition 1.9 explicitly writes only the second condition. Domain stationarity alone is not the full hypothesis of the later harmonic-map regularity theorems. Why is it unsafe to substitute  $V = du(X)$  into a weak first variation? Its weak derivative would require second derivatives of  $u$ , which  $W^{1,2}$  does not supply.

## 5 Stationary identity and monotonicity

**Source:** §1.5.1–1.5.2, pp. 12–14; equations (1.10)–(1.11); Theorem 2.1, p. 15.

### Stationary identity.

For a stationary map and every compactly supported vector field  $X$ ,

$$\int_M \left( |du|^2 \operatorname{div} X - 2 \sum_{a,b} \langle du(e_a), du(e_b) \rangle \langle \nabla_{e_a} X, e_b \rangle \right) dV_g = 0.$$

**Proof idea.** Change variables in  $E(u \circ \phi_s, g)$  and use  $E(u \circ \phi_s, g) = E(u, \phi_{-s}^* g)$ . Differentiate the inverse metric and volume factor. Their contributions are the two terms above.

This is conservation of the stress tensor  $S = \frac{1}{2}|du|^2 g - u^* h$ , expressed weakly. It needs only first derivatives of  $u$ , so it is meaningful for Sobolev maps.

### Monotonicity formula in a Euclidean domain.

Let  $u \in W^{1,2}(\Omega, N)$  be stationary,  $\Omega \subset \mathbb{R}^m$ , and  $B_R(x_0) \Subset \Omega$ . Set

$$\Theta_u(x_0, r) = r^{2-m} \int_{B_r(x_0)} |\nabla u|^2 dx.$$

Then for  $0 < r < R$ ,

$$\Theta_u(x_0, R) - \Theta_u(x_0, r) = 2 \int_{B_R(x_0) \setminus B_r(x_0)} |x - x_0|^{2-m} |\partial_\rho u|^2 dx \geq 0.$$

In differential form, for almost every  $r$ ,

$$\frac{d}{dr} \Theta_u(x_0, r) = 2r^{2-m} \int_{\partial B_r(x_0)} |\partial_r u|^2 dS.$$

**Proof idea.** Insert  $X(x) = \eta(|x - x_0|)(x - x_0)$  into the stationary identity. Writing  $\rho = |x - x_0|$  yields

$$0 = \int ((m-2)\eta + \rho\eta') |\nabla u|^2 - 2 \int \rho\eta' |\partial_\rho u|^2.$$

Approximate a ball indicator by radial cutoffs, apply the coarea formula, and rearrange the derivative of  $r^{2-m} \int_{B_r} |\nabla u|^2$ .

The power  $2 - m$  is forced by scaling:

$$v(y) = u(x_0 + ry) \implies \int_{B_1} |\nabla v|^2 = r^{2-m} \int_{B_r(x_0)} |\nabla u|^2.$$

If normalized energy is constant through an interval of radii, its radial derivative vanishes almost everywhere there. This is the route from monotonicity to homogeneous tangent maps. The exact identity above uses the Euclidean metric. A curved domain has metric error terms and a corresponding local almost-monotonicity formula. Also, the factor 2 is missing in the display in §2.6.1 on p. 28; it is present correctly in (1.11).

## 6 Small-energy regularity for smooth harmonic maps

**Source:** §2.1, pp. 15–17; Lemma 2.2 and Theorem 2.3.

### Lemma 2.2.

For a smooth harmonic map on a region with controlled geometry,

$$\Delta |\nabla u|^2 \geq -C(|\nabla u|^2 + |\nabla u|^4).$$

This follows by bounding the curvature terms in the Bochner identity.

### Theorem 2.3: Smooth $\varepsilon$ -regularity.

There exist  $\varepsilon > 0$  and  $C$ , depending on the controlled domain and target geometry, such that a smooth harmonic map on  $B_r$  satisfies

$$r^{2-m} \int_{B_r} e(u) \leq \varepsilon \implies \boxed{\sup_{B_{r/2}} e(u) \leq Cr^{-m} \int_{B_r} e(u)}.$$

The source assumes  $u \in C^2$ ; Proposition 1.6 then gives smoothness.

The small quantity is *scale-invariant* energy. In dimension  $m > 2$ , a small raw integral on a tiny ball by itself is not the correct hypothesis. The estimate rules out a large gradient spike when normalized energy is small.

**Proof idea.** Use a point-picking and rescaling argument.

1. Maximize  $(3r/4 - \sigma)^2 \sup_{B_\sigma} e(u)$  over  $0 \leq \sigma \leq 3r/4$ . At an almost-maximizing point  $x_0$ , write  $e_0 = e(u)(x_0)$  and choose  $\rho_0$  comparable to the remaining distance. Then  $e(u) \leq 4e_0$  on  $B_{\rho_0}(x_0)$ .
2. Rescale  $v(y) = u(x_0 + y/\sqrt{e_0})$ . Now  $e(v)(0) = 1$  and  $e(v) \leq 4$  on the corresponding ball. The Bochner inequality becomes a linear subsolution inequality  $\Delta e(v) \geq -Ce(v)$ .
3. A mean-value estimate bounds  $e(v)(0)$  by an integral of  $e(v)$ . If the rescaled ball contains a unit ball, monotonicity bounds this integral by  $C\varepsilon$ , contradicting  $e(v)(0) = 1$  for small  $\varepsilon$ .
4. Apply the same estimate at the controlled scale and undo the rescaling. The maximal weighted energy density bounds the whole inner ball.

Once  $|\nabla u|$  is bounded, the equation has bounded right-hand side. Interior elliptic estimates give  $W^{2,p}$  and then higher regularity. This is how a first-derivative estimate controls the entire smooth map on smaller balls. Despite the section title, Theorem 2.3 is stated for *smooth harmonic maps*, without a minimizing hypothesis. Its proof does not apply directly to an initially nonsmooth  $W^{1,2}$  map: the pointwise Bochner argument would be circular.

## 7 Stationary regularity and the conservation-law structure

**Source:** §2.4, pp. 20–24; Lemma 2.7, Definitions 2.8, 2.11–2.12, Theorems 2.9–2.10 and 2.13–2.14.

The green sentence on p. 20 records that small-energy regularity extends to general stationary harmonic maps. The source explains the sphere-target case  $u : B_3^m \rightarrow S^{L-1}$ , where

$$\Delta u + |\nabla u|^2 u = 0.$$

### 7.1 Antisymmetric currents

**Lemma 2.7, with a consistent sign.**

Define

$$V^{ij} = u^j \nabla u^i - u^i \nabla u^j.$$

Then  $\operatorname{div} V^{ij} = 0$  distributionally, and

$$\Delta u^i = \sum_j \nabla u^j \cdot V^{ij}.$$

**Proof idea.** The divergence is  $u^j \Delta u^i - u^i \Delta u^j = 0$  by the sphere equation. The identity  $\sum_j u^j \nabla u^j = 0$  gives  $\sum_j \nabla u^j \cdot V^{ij} = -u^i |\nabla u|^2 = \Delta u^i$ . For weak maps, these statements follow by distributional testing with admissible products, rather than a pointwise second-derivative calculation. The source has  $-\Delta u^i$  with this definition on p. 21, but  $\Delta u^i$  on p. 23. The latter is consistent. Reversing  $V^{ij}$  would give an equally valid opposite-sign formulation.

### 7.2 The functional spaces used by the proof

**Hardy, Morrey, and BMO quantities.**

For a standard smooth approximate identity  $\eta_t$ ,

$$f^*(x) = \sup_{t>0} |\eta_t * f(x)|, \quad \|f\|_{\mathcal{H}^1} = \|f\|_{L^1} + \|f^*\|_{L^1}.$$

The source's Morrey seminorm is

$$\|f\|_{M^{p,m-p}(U)}^p = \sup_{B_r(x) \subset U} r^{p-m} \int_{B_r(x)} |f|^p.$$

For  $f_B = |B|^{-1} \int_B f$ ,

$$[f]_{\operatorname{BMO}(U)} = \sup_{B \subset U} \frac{1}{|B|} \int_B |f - f_B|.$$

The source uses  $r^{-m}$  instead of  $|B_r|^{-1}$  in BMO; this differs only by a dimensional constant.

**Theorems 2.9, 2.10, and 2.13: the analytical tools.**

For the Newtonian solution of  $-\Delta \varphi = f$  on  $\mathbb{R}^m$ ,

$$\|D^2 \varphi\|_{L^1} \leq C \|f\|_{\mathcal{H}^1}.$$

If  $1 < p < \infty$ ,  $q = p/(p-1)$ , and  $G \in L^q$  satisfies  $\operatorname{div} G = 0$ , then

$$\|\nabla h \cdot G\|_{\mathcal{H}^1} \leq C \|\nabla h\|_{L^p} \|G\|_{L^q}.$$

Hardy–BMO duality consequently gives, for  $p = q = 2$ ,

$$\left| \int_{\mathbb{R}^m} (\nabla h \cdot G) b \right| \leq C \|\nabla h\|_2 \|G\|_2 [b]_{\text{BMO}}.$$

A generic product of two  $L^2$  fields is only in  $L^1$ . The divergence-free condition supplies additional cancellation. It allows the nonlinear equation to be estimated using oscillation of  $u$ , instead of requiring a pointwise bound on its gradient. The divergence-free hypothesis is essential and is missing from the statement of Theorem 2.10 on p. 21. The Newtonian-solution qualification in Theorem 2.9 excludes arbitrary additions of harmonic functions with uncontrolled Hessian.

### 7.3 The energy-decay mechanism

**Theorem 2.14, with  $m$  used consistently for domain dimension.**

There are  $\varepsilon_0 > 0$  and  $\theta_0 \in (0, 1/2)$  such that, if  $u$  is stationary harmonic on  $B_3^m$  and  $\int_{B_3} |\nabla u|^2 \leq \varepsilon_0^2$ , then for  $z \in B_{1/2}$ ,

$$\theta_0^{2-m} \int_{B_{\theta_0}(z)} |\nabla u|^2 \leq \frac{1}{4} \int_{B_1} |\nabla u|^2.$$

Iterating at smaller scales gives Hölder continuity and then smoothness.

**Proof idea.** Monotonicity and Poincaré imply  $[u]_{\text{BMO}} \leq C\varepsilon_0$ . Compare  $u$  with an ambient harmonic function  $v$  having the same boundary values on a ball. Test the equation against  $u - v$ , use  $\text{div } V^{ij} = 0$ , and apply the Hardy–BMO estimate to obtain

$$\int |\nabla(u - v)|^2 \leq C\varepsilon_0^2 \int |\nabla u|^2.$$

Harmonic functions have quadratic decay of normalized energy. Thus the decay factor for  $u$  is bounded by

$$C\varepsilon_0^2 \theta^{2-m} + C\theta^2.$$

Choose  $\theta$  small first, then  $\varepsilon_0$  small. Iteration yields  $r^{2-m} \int_{B_r} |\nabla u|^2 \leq Cr^{2\alpha}$ , and Morrey’s lemma gives  $C^\alpha$  regularity. The upgrade to smoothness uses the regularity principle discussed in the unmarked §2.3.

## 8 Density, singular sets, and minimizing maps

**Source:** §2.6, pp. 25–29; Propositions 2.17 and 2.19; Theorems 2.18, 2.20–2.21; Corollary 2.22.

### 8.1 Density detects singularities

**Density and the regular set.**

In the Euclidean setting define

$$\Theta_u(x) = \lim_{r \downarrow 0} r^{2-m} \int_{B_r(x)} |\nabla u|^2.$$

Monotonicity gives existence. At a smooth point it is zero, since  $\int_{B_r} |\nabla u|^2 = O(r^m)$ . Small-energy regularity gives the converse and a fixed positive density gap at singular points. Thus

$$\text{sing } u = \{x : \Theta_u(x) > 0\}, \quad \Theta_u(x) \geq c_* > 0 \quad (x \in \text{sing } u).$$

The density is upper semicontinuous, so the singular set is relatively closed. The exact letter and square used for the small threshold vary in the source;  $c_*$  avoids identifying two differently normalized constants.

**Proposition 2.17.**

For a stationary harmonic map,

$$\mathcal{H}^{m-2}(\text{sing } u) = 0.$$

**Proof idea.** At a singular point every sufficiently small ball contains at least  $c_* r^{m-2}$  energy. Cover a compact part of the singular set by small balls with bounded overlap. First the estimate implies zero Lebesgue measure. Then absolute continuity of the integral gives

$$\sum_i r_i^{m-2} \leq C \int_{\text{shrinking neighborhood}} |\nabla u|^2 \longrightarrow 0.$$

This is exactly the Hausdorff measure conclusion. This theorem says more than  $\dim_H \text{sing } u \leq m-2$ , but it does not automatically imply  $\dim_H \text{sing } u \leq m-3$ . Nor does a zero-dimensional set automatically have isolated points.

**8.2 The radial minimizing example****Proposition 2.19.**

For  $m \geq 3$ ,

$$\phi_0(x) = \frac{x}{|x|} : B^m \setminus \{0\} \rightarrow S^{m-1}$$

defines a  $W^{1,2}$  energy minimizer with identity boundary values. Its central singularity is genuine, and

$$|\nabla \phi_0|^2 = \frac{m-1}{|x|^2}, \quad \mathcal{E}(\phi_0; B_1) = \frac{m-1}{m-2} |S^{m-1}|.$$

**Proof idea.** The source uses the inequality, for a sphere-valued competitor,

$$|\nabla \phi|^2 \geq \frac{1}{m-2} ((\text{div } \phi)^2 - \text{tr}((\nabla \phi)^2)).$$

The expression in parentheses is the divergence of  $(\text{div } \phi)\phi - (\nabla \phi)\phi$ . Its boundary flux for identity boundary data is  $(m-1)|S^{m-1}|$ . Integrating yields the lower bound attained by  $\phi_0$ . For Sobolev competitors the flux identity must be justified weakly, or by approximation and the null-Lagrangian structure; normal derivatives are not ordinary boundary traces. The singularity costs finite energy because  $\int_0^1 r^{m-3} dr < \infty$  precisely when  $m > 2$ . The map is homogeneous of degree zero, so normalized energy is constant in radius. This is a concrete model for a minimizing tangent map.

**8.3 Sharper regularity for minimizers****Theorems 2.18 and 2.20.**

An energy-minimizing map is smooth away from a relatively closed set, and

$$\boxed{\dim_H \text{sing } u \leq m-3 \quad (m \geq 4).}$$

For  $m = 3$ , its interior singularities are isolated.

**Theorem 2.21: decay for a minimizing map.**

For some  $\varepsilon_0 > 0$  and  $\theta_0 \in (0, 1/2)$ ,

$$\int_{B_1} |\nabla u|^2 \leq \varepsilon_0^2 \implies \theta_0^{2-m} \int_{B_{\theta_0}} |\nabla u|^2 \leq \frac{1}{2} \int_{B_1} |\nabla u|^2.$$

After applying the estimate on translated and rescaled balls,

$$r^{2-m} \int_{B_r(x)} |\nabla u|^2 \leq C r^{2\alpha}, \quad u \in C_{\text{loc}}^\alpha,$$

and the harmonic-map equation upgrades regularity to smoothness.

**Proof idea.** Normalize a hypothetical failure of decay around almost-constant maps. The limiting linear problem is a harmonic function with values in a tangent space of  $N$ , whose energy has better decay. Minimality is used to pass the energy comparison to the limit. This is the compactness and harmonic-approximation mechanism behind the stated estimate.

## 8.4 Why the minimizing hypothesis improves the dimension bound

**Source:** §2.6.3, pp. 30–34; Lemma 2.23, Definition 2.24, Proposition 2.25, Definition 2.26, and Theorem 2.27

Rescale  $u_{x,r}(y) = u(x + ry)$ . Strong local  $W^{1,2}$  compactness of minimizers produces a tangent map  $\phi$  as  $r \downarrow 0$ . No energy is lost in this strong limit. Consequently

$$\rho^{2-m} \int_{B_\rho} |\nabla \phi|^2 = \Theta_u(x) \quad \text{for every } \rho > 0.$$

Equality in monotonicity makes  $\phi$  homogeneous of degree zero. Its maximal-density set

$$S(\phi) = \{y : \Theta_\phi(y) = \Theta_\phi(0)\}$$

is a linear space of translation invariance. A nonconstant tangent map cannot have an  $(m-2)$ -dimensional such space, because this would violate  $\mathcal{H}^{m-2}(\text{sing } \phi) = 0$ .

Dimension reduction converts these restrictions on tangent symmetries into the bound  $m-3$ . The same proof cannot simply be copied for stationary maps: weak convergence may carry an energy defect, so the limiting map need not retain all of the rescaled energy.

For  $m=3$ , isolatedness needs an additional compactness and dimension-reduction argument; it is not a consequence of the numerical bound  $\dim_H \text{sing } u = 0$  alone. An accumulating sequence of singular points would, after rescaling, give a tangent map with a nonzero singular point. Homogeneity would then produce a singular ray, contradicting the zero  $\mathcal{H}^1$  measure conclusion.

## 8.5 Step 3: stratification of the singular set

**Definition 2.26 and Theorem 2.27.**

For  $0 \leq j \leq m-3$ , define

$$S_j(u) = \{x \in \text{sing } u : \dim S(\phi) \leq j \text{ for every tangent map } \phi \text{ at } x\}.$$

Then

$$S_0(u) \subset S_1(u) \subset \cdots \subset S_{m-3}(u) = \text{sing } u, \quad \boxed{\dim_H S_j(u) \leq j}.$$

The quantifier “every tangent map” matters. A point belongs to  $S_j$  if no blow-up at that point has more than  $j$  independent translation symmetries. The theorem links the dimension of a set of singular points to the symmetries of its local models.

**Proof idea.** The source organizes the proof into approximation, decomposition, and covering.

1. **Near-maximal-density points lie near a plane.** Fix  $x \in S_j$  and  $\delta > 0$ . On sufficiently small scales  $\rho$ , the rescaled set of points  $y \in B_\rho(x)$  satisfying  $\Theta_u(y) \geq \Theta_u(x) - \varepsilon$  lies in the  $\delta$ -neighborhood of some  $j$ -plane. Otherwise take a strongly convergent blow-up sequence. Upper semicontinuity of density puts each limiting near-maximal point in  $S(\phi)$ , whose dimension is at most  $j$ , a contradiction. The plane must approximate the *whole set*; selecting a single point outside every possible  $j$ -plane would not be a valid formulation.
2. **Make the scales and density tolerances uniform.** Decompose  $S_j$  countably according to a lower bound on the available scale and short intervals of density values. Within each piece, points have comparable densities, so the approximation property applies uniformly at every sufficiently small ball centered in that piece.
3. **Cover and iterate.** A  $\delta\rho$ -neighborhood of a  $j$ -plane inside  $B_\rho$  is covered by at most  $C\delta^{-j}$  balls of radius comparable to  $\delta\rho$ . Iterate the covering. For every  $s > j$ , choose  $\delta$  small enough that the  $s$ -dimensional covering content contracts. Each piece has Hausdorff dimension at most  $s$ ; countable unions and then  $s \downarrow j$  give the result.

In the last step, the countable decomposition may depend on  $\delta$ . Fix  $s > j$ , choose  $\delta$ , and estimate that decomposition before taking the countable union. This avoids silently treating  $\delta$ -dependent pieces as fixed sets when sending  $\delta \rightarrow 0$ .

## 9 Surfaces, conformal invariance, and the Hopf differential

**Source:** §4.1, pp. 41–44; Propositions 4.1–4.3, Definition 4.4, Lemmas 4.5–4.6, Corollary 4.7.

### 9.1 Why dimension two is special

**Conformal invariance and Proposition 4.1.**

If  $\dim \Sigma = 2$  and  $g' = e^{2\omega}g$ , then

$$|du|_{g'}^2 = e^{-2\omega}|du|_g^2, \quad dV_{g'} = e^{2\omega} dV_g, \quad E(u, g') = E(u, g).$$

If  $f : (\Sigma, g_1) \rightarrow (\Sigma, g_2)$  is a conformal diffeomorphism and  $u : (\Sigma, g_2) \rightarrow N$  is harmonic, then  $u \circ f$  is harmonic.

**Proof idea.** The two conformal factors cancel in the integrand. Transport each variation of  $u \circ f$  by  $f^{-1}$  and apply criticality of  $u$ . In dimension  $m$ , the net factor is  $e^{(m-2)\omega}$ . Its disappearance at  $m = 2$  explains both conformal invariance and the scale-invariant energy concentration underlying bubbling.

**Proposition 4.2: Area–energy inequality (normalized).**

Let  $A(u)$  be parametrized area, counted with multiplicity. Then

$$A(u) \leq E(u),$$

with equality precisely when  $u$  is weakly conformal:  $u^*h = \lambda g$ ,  $\lambda \geq 0$ .

**Proof idea.** If  $\lambda_1, \lambda_2 \geq 0$  are eigenvalues of  $g^{-1}u^*h$ , the energy and area densities are respectively  $\frac{1}{2}(\lambda_1 + \lambda_2)$  and  $\sqrt{\lambda_1\lambda_2}$ . Apply the arithmetic–geometric mean inequality. The source writes  $A \leq \frac{1}{2}E$ , using its later unhalved energy convention. Here  $A \leq E = \frac{1}{2}\mathcal{E}$ .

**Proposition 4.3.**

A weakly conformal harmonic map from a surface is stationary for area (and is a minimal immersion wherever its differential has rank two).

**Proof idea.** At the map itself,  $A = E$ , while along any variation  $A(u_t) \leq E(u_t)$ . First derivatives in opposite directions show  $\delta A = 0$ . “Minimal” here means an area critical point, not necessarily an area minimizer. Branch points, where  $du = 0$ , are allowed; a nonconstant harmonic sphere is understood in this branched sense.

### 9.2 The Hopf differential

**Definition 4.4 and Lemma 4.5.**

In a conformal coordinate  $z = x + iy$ , set

$$\varphi(z) = |u_x|^2 - |u_y|^2 - 2i\langle u_x, u_y \rangle, \quad \Phi = \varphi(z) dz^2.$$

It is a globally defined quadratic differential. If  $u$  is harmonic,  $\Phi$  is holomorphic.

**Proof idea.** With  $u_z = \frac{1}{2}(u_x - iu_y)$  and the complex bilinear target metric,  $\varphi = 4(u_z, u_z)$ . Harmonicity is  $\nabla_{\bar{z}}u_z = 0$ , so

$$\partial_{\bar{z}}\varphi = 8(\nabla_{\bar{z}}u_z, u_z) = 0.$$

The quadratic transformation law under a holomorphic coordinate change makes  $\Phi$  independent of the chosen conformal chart. The real part measures unequal stretching in the  $x$  and  $y$  directions. The imaginary part measures failure of orthogonality. Thus

$$\Phi = 0 \iff |u_x| = |u_y|, \quad \langle u_x, u_y \rangle = 0 \iff u \text{ is weakly conformal.}$$

This is Lemma 4.6; the algebraic equivalence itself does not require harmonicity.

**Corollary 4.7: Harmonic spheres.**

Every harmonic map  $u : S^2 \rightarrow N$  has  $\Phi(u) = 0$ . Hence a nonconstant harmonic sphere is weakly conformal and minimal in the branched sense.

**Proof idea.** Stereographic projection gives a harmonic map on  $\mathbb{C}$  with finite energy. Its Hopf coefficient is entire and satisfies  $\int_{\mathbb{C}} |\varphi| \leq CE(u) < \infty$ . The mean-value inequality gives  $|\varphi(z_0)| \leq (\pi R^2)^{-1} \int_{\mathbb{C}} |\varphi| \rightarrow 0$  as  $R \rightarrow \infty$ . Thus  $\varphi \equiv 0$ . A harmonic map from a higher-genus surface need not be conformal: that surface can carry a nonzero holomorphic quadratic differential.

## 10 Sacks–Uhlenbeck theory and the existence of harmonic spheres

**Source:** §4.2, p. 44; Theorem 4.8 and Proposition 4.9. The roadmap below refers to §4.2.1–4.2.3, pp. 44–51.

**Theorem 4.8 (Sacks–Uhlenbeck).**

For a closed Riemannian target  $(N, h)$ , if

$$\pi_k(N) \neq 0 \quad \text{for some } k \geq 2,$$

then there exists a nonconstant harmonic map  $u : S^2 \rightarrow N$ . Topological information in a possibly higher homotopy group forces a two-dimensional geometric object. By Corollary 4.7, the resulting sphere is also a branched minimal sphere. “Nontrivial” here means nonconstant, and does not assert that every such sphere is a minimizer or represents a prescribed homotopy class.

**The  $\alpha$ -energy and its Euler–Lagrange equation.**

For  $\alpha > 1$  on a closed surface,

$$E_\alpha(u) = \int_{\Sigma} (1 + |\nabla u|^2)^\alpha dV_g.$$

Writing  $q = 1 + |\nabla u|^2$ , critical maps satisfy

$$\boxed{\operatorname{div}(q^{\alpha-1} \nabla u) + q^{\alpha-1} A(u)(\nabla u, \nabla u) = 0,}$$

or, componentwise,

$$\Delta u^i + (\alpha - 1) \frac{\langle \nabla |\nabla u|^2, \nabla u^i \rangle}{1 + |\nabla u|^2} + A^i(u)(\nabla u, \nabla u) = 0.$$

**Proposition 4.9.**

Critical points of  $E_\alpha$  in

Since  $2\alpha > 2$ , the relevant Sobolev maps are continuous and the problem lies above the critical exponent. As  $\alpha \downarrow 1$ , the equation approaches the harmonic map equation, but compactness can fail by concentration. Notice also  $E_1(u) = \operatorname{vol}(\Sigma) + 2E(u)$  in our convention.

## 10.1 Supplementary proof roadmap: estimates, removal, bubbles

**Supplement (not independently green-marked).** The following supporting results are in the unmarked child subsections. They explain how the marked existence theorem works.

### Local estimate (Theorem 4.10).

For sufficiently small Dirichlet energy, a critical point of  $E_\alpha$  on a disk satisfies an interior estimate of the form

$$\|\nabla u\|_{W^{1,p}(D_{1/2})} \leq C(p) \|\nabla u\|_{L^2(D)},$$

for fixed finite  $p$  and  $\alpha > 1$  sufficiently close to one.

**Proof idea.** Localize the equation. The perturbation term is controlled by  $C(\alpha - 1)|D^2u|$  and is absorbed into the elliptic estimate. The quadratic term is absorbed using small energy and, initially,  $W^{2,4/3} \hookrightarrow W^{1,4}$  in two dimensions. Bootstrap on smaller disks. The permitted parameters may depend on  $p$ ; no uniform estimate as  $p \rightarrow \infty$  is being asserted.

### Energy gap (Theorem 4.11, domain qualification).

On a fixed *closed* domain surface, sufficiently small energy forces an  $E_\alpha$ -critical map to be constant, for  $\alpha$  close to one. This is not true for arbitrary maps on a disk without boundary conditions.

Small-energy estimates put the image in a small target neighborhood. Testing against an appropriate displacement and using Poincaré or a strictly convex local target function forces zero energy. On a disk, short nonconstant geodesic-valued harmonic maps give immediate counterexamples to an unqualified gap statement.

### Removal of a point singularity (Theorem 4.12).

A smooth harmonic map  $u : D \setminus \{0\} \rightarrow N$  with finite energy extends smoothly and harmonically across the origin.

**Proof idea.** Use the conformal cylinder  $t = \log r$ . The quantity  $\int_{S^1} (|u_t|^2 - |u_\theta|^2) d\theta$  is constant in  $t$ ; finite energy makes that constant zero. Thus

$$r^2 \int_{S^1} |u_r|^2 d\theta = \int_{S^1} |u_\theta|^2 d\theta.$$

Compare with circle averages and absorb the small nonlinear error. For  $F(r) = \int_{D_r} |\nabla u|^2$ , the resulting estimate is

$$rF'(r) \geq \beta F(r), \quad F(r) \leq Cr^\beta, \quad \beta = 1 - 2\delta > 0.$$

Local gradient estimates then give  $|\nabla u(x)| \leq C|x|^{-1+\beta/2}$ , which belongs to  $L^p$  for some  $p > 2$ . The point has zero  $W^{1,2}$  capacity, so the weak equation extends across it; continuity and elliptic regularity finish the proof.

The gradient exponent written on p. 50 does not follow from the displayed energy decay. The exponent  $-1 + \beta/2$  above is the one obtained by applying the scaled interior estimate; it still gives the required  $p > 2$ .

### Bubbling mechanism (§4.2.3).

For a suitable bounded-energy sequence of  $E_{\alpha_i}$ -critical maps with  $\alpha_i \downarrow 1$ , either there is smooth convergence, or energy concentrates at finitely many points. Away from those points, small-energy estimates give a smooth harmonic limit. Rescaling at a concentration scale produces a nonconstant finite-energy harmonic map on  $\mathbb{R}^2$ , and removal at infinity yields a harmonic sphere. The original limit may be constant while a rescaled bubble is nonconstant. A positive energy threshold bounds the number of concentration points. This roadmap explains the compactness

alternative; it does not assert a full energy identity or a no-neck theorem.

The construction of the required minimization or min-max sequence and the full rescaling argument are substantial parts of the theory. The source only sketches them here. This outline is not a replacement for their complete proofs.

## 11 Positive isotropic curvature

**Source:** §5.1, pp. 55–56; Definitions 5.1–5.2. Supporting algebra from pp. 57–59 is labeled below.

To match Chapter 5 of the source, write the target as  $(M^n, g)$  in the remaining sections. Complicate its tangent space:  $T_p M \otimes_{\mathbb{R}} \mathbb{C}$ . There are two distinct metric extensions:

$$(z, w) = \text{complex bilinear extension}, \quad \langle\langle z, w \rangle\rangle = (z, \bar{w}) = \text{Hermitian extension}.$$

### Definition 5.1.

A vector  $z$  is isotropic if  $(z, z) = 0$ . A complex linear subspace  $W$  is totally isotropic if every  $z \in W$  is isotropic. By polarization, this is equivalent to  $(z, w) = 0$  for all  $z, w \in W$ .

For  $z = a + ib$ ,

$$(z, z) = |a|^2 - |b|^2 + 2i\langle a, b \rangle.$$

Thus isotropy means  $|a| = |b|$  and  $a \perp b$ , not  $z = 0$ . For example  $e_1 + ie_2$  is isotropic but has Hermitian squared norm 2. Confusing the two metric extensions destroys the definition.

### Maximum isotropic dimension.

For a totally isotropic subspace of  $\mathbb{C}^n$ ,

$$\dim_{\mathbb{C}} W \leq \left\lfloor \frac{n}{2} \right\rfloor.$$

**Proof idea.** Bilinear orthogonality gives  $W \subset W^\perp$ . Nondegeneracy gives  $\dim W + \dim W^\perp = n$ , so  $2 \dim W \leq n$ . The span of  $e_1 + ie_2, e_3 + ie_4, \dots$  attains the bound.

### Definition 5.2: Positive isotropic curvature (PIC).

Let  $\mathcal{R} : \Lambda^2 T_p M \rightarrow \Lambda^2 T_p M$  be the curvature operator, extended complex linearly. For a complex two-plane  $\sigma = \text{span}_{\mathbb{C}}\{z, w\}$ , define

$$K_{\mathbb{C}}(\sigma) = \frac{\langle\langle \mathcal{R}(z \wedge w), z \wedge w \rangle\rangle}{\|z \wedge w\|^2}.$$

The manifold has PIC if  $K_{\mathbb{C}}(\sigma) > 0$  on every totally isotropic complex two-plane at every point. This condition is nonvacuous for  $n \geq 4$ .

**Supplement (not independently green-marked).** A totally isotropic two-plane has a basis  $z = e_1 + ie_2, w = e_3 + ie_4$  for a real orthonormal four-frame (Lemma 5.3). Expanding  $z \wedge w$  and using Bianchi gives equation (5.1):

$$\mathcal{I}(e_1, e_2, e_3, e_4) = R_{1313} + R_{1414} + R_{2323} + R_{2424} - 2R_{1234}.$$

Here  $R_{ijkl} = \text{Rm}(e_i, e_j, e_k, e_l)$  and  $\|z \wedge w\|^2 = 4$ . PIC is equivalent to  $\mathcal{I} > 0$  for every orthonormal four-frame. The normalization of  $z, w$  does not affect the sign.

This converts a complex-geometric positivity condition into an explicit real curvature inequality. Positivity of the entire real curvature operator implies PIC, since the Hermitian expression is the sum of its values on the real and imaginary parts of  $z \wedge w$  (Proposition 5.4, a supporting unmarked result).

## 12 Pointwise quarter-pinching implies PIC

**Source:** §5.1.5, pp. 59–60; Definition 5.5 and Proposition 5.6. Berger’s inequality is the supporting Lemma 5.7, pp. 60–62.

**Definition 5.5: Pointwise  $\delta$ -pinching.**

For  $0 < \delta \leq 1$ , there is a function  $k(p) > 0$  such that

$$\delta k(p) \leq K(\sigma) \leq k(p)$$

for every real two-plane  $\sigma \subset T_p M$ . It is global pinching if  $k$  is constant. Strict quarter-pinching can equivalently be phrased pointwise as  $0 < b(p) < 4a(p)$ , where  $a(p)$  and  $b(p)$  are the minimum and maximum sectional curvatures.

**Proposition 5.6.**

A strictly pointwise  $1/4$ -pinched manifold has positive isotropic curvature.

**Supplement (not independently green-marked).** Berger’s inequality says that  $a \leq K \leq b$  implies

$$|R_{1234}| \leq \frac{2}{3}(b - a)$$

on every orthonormal four-frame. Polarize sectional-curvature expressions and use the curvature symmetries and Bianchi identity to isolate the mixed component. An intermediate estimate is  $|R(u, v, w, v)| \leq \frac{1}{2}(b - a)$  for orthonormal  $u, v, w$ .

**Proof idea.** At a fixed point,

$$\begin{aligned} \mathcal{I} &\geq 4a - 2|R_{1234}| \\ &\geq 4a - \frac{4}{3}(b - a) = \frac{4}{3}(4a - b) > 0. \end{aligned}$$

Strictness matters. Non-strict quarter-pinching yields only  $\mathcal{I} \geq 0$  by this argument. The assumption is pointwise, so the curvature scale may vary from point to point.

Locate the exact use of the number  $1/4$ : the four positive sectional-curvature terms must dominate twice a mixed curvature term bounded by  $\frac{2}{3}(b - a)$ .

## 13 The sphere theorem and the index of harmonic two-spheres

**Source:** §5.2 and §5.2.1, pp. 62–64; Theorems 5.8–5.9 and Lemma 5.10.

**Theorem 5.8: the sphere-theorem consequence in the text.**

Let  $(M^n, g)$  be *closed*, simply connected, and  $n \geq 4$ . If  $M$  is strictly pointwise  $1/4$ -pinched, then it is homeomorphic to  $S^n$ .

Closedness is made explicit here. It is necessary and is implicit in the geometric setting; the isolated display on p. 62 omits it. The conclusion stated in the source is a homeomorphism, not an isometry or a claim about the differentiable structure.

**Theorem 5.9: index bound under PIC.**

If  $M^n$  has positive isotropic curvature, every nonconstant conformal harmonic map  $f : S^2 \rightarrow M$  satisfies

$$\boxed{\text{ind}(f) \geq \frac{n-3}{2}.}$$

Since the index is an integer, this means  $\text{ind}(f) \geq \lceil (n-3)/2 \rceil$ .

PIC forces enough negative directions for the energy Hessian. A nonconstant sphere cannot

be too stable. The sphere theorem combines this restriction with variational constructions of low-index spheres from topology. Those constructions are discussed later in unmarked §5.3; they are not proved by the index inequality alone.

### 13.1 The holomorphic pullback bundle

For a harmonic map from a Riemann surface, define

$$\mathcal{E} = f^*TM \otimes \mathbb{C}, \quad \nabla = \nabla' + \nabla'', \quad \bar{\partial}_{\mathcal{E}} = \nabla''.$$

On a complex curve the  $(0,2)$  obstruction vanishes, so this operator defines a holomorphic structure. A section  $W$  is holomorphic exactly when  $\nabla_{\bar{z}}W = 0$ .

**Lemma 5.10 and the conformality criterion.**

Locally put  $f_z = \frac{1}{2}(f_x - if_y)$ . Then

$$f \text{ harmonic} \iff \nabla_{\bar{z}}f_z = 0, \quad f \text{ weakly conformal} \iff (f_z, f_z) = 0.$$

**Proof idea.** Torsion-freeness gives  $\nabla_x f_y = \nabla_y f_x$ , hence

$$\nabla_{\bar{z}}f_z = \frac{1}{4}(\nabla_x f_x + \nabla_y f_y).$$

For  $g = \lambda^2 |dz|^2$ , the right-hand side is  $\lambda^2 \tau(f)/4$ . Also  $4(f_z, f_z) = |f_x|^2 - |f_y|^2 - 2i\langle f_x, f_y \rangle$ .

A conformal harmonic map produces a holomorphic isotropic object. This is precisely the kind of object on which PIC can be tested.

Globally  $f_z dz$  is an  $\mathcal{E}$ -valued  $(1,0)$ -form, or a section of  $K_{\Sigma} \otimes \mathcal{E}$ . The coefficient  $f_z$  alone is chart-dependent. Statements about global spaces of sections must respect this distinction and possible branch points.

## 14 Second variation, Jacobi fields, and complex index form

**Source:** §5.2.2, pp. 64–66; Definition 5.11 and Lemma 5.12, equation (5.2).

**Real index form, with the energy normalization fixed.**

For real sections  $V, W \in \Gamma(f^*TM)$ , set

$$I(V, W) = \delta^2 E_f(V, W) = \int_{\Sigma} \left( \langle \nabla V, \nabla W \rangle - \sum_a \text{Rm}(V, df(e_a), W, df(e_a)) \right) dA.$$

Here  $e_a$  is an orthonormal frame on the *domain*; its image  $df(e_a)$  enters the target curvature tensor.

**Definition 5.11: Index, nullity, Jacobi field.**

The index is the largest dimension of a real vector space of sections on which  $I$  is negative definite. A Jacobi field  $V$  satisfies

$$I(V, W) = 0 \quad \text{for every smooth } W.$$

The dimension of this kernel is the nullity. On a closed domain, integration by parts expresses  $I$  through a self-adjoint elliptic Jacobi operator, so the index counts its negative eigenvalues with multiplicity, using the corresponding Hessian sign convention.

Negative directions lower energy to second order. Kernel directions have no quadratic energy change and may arise from symmetries. A vector with  $I(V, V) = 0$  is not necessarily a Jacobi field: vanishing against *every* test section is the stronger condition.

**Lemma 5.12: complex form of the second variation.**

Extend  $I$  sesquilinearly to the complexified bundle. With our normalization,

$$I_{\mathbb{C}}(W, W) = 4 \int_{\Sigma} \left( \|\nabla_{\bar{z}} W\|^2 - \langle\langle \mathcal{R}(W \wedge f_z), W \wedge f_z \rangle\rangle \right) dx dy.$$

The coordinate densities combine to a globally defined integral. The source has coefficient 8, corresponding to  $\mathcal{E} = 2E$ ; multiplying the whole Hessian by a positive constant does not change index or nullity.

**Proof idea.** Expand  $\nabla_{\bar{z}} = \frac{1}{2}(\nabla_x + i\nabla_y)$ . The mixed derivative terms become curvature after integration by parts and the commutator identity  $[\nabla_x, \nabla_y]W = R(f_x, f_y)W$ . Combine them with the real curvature term using Bianchi. The result is the curvature of the complex two-plane spanned by  $W$  and  $f_z$ .

If  $W$  is holomorphic, the first term vanishes. If  $\text{span}_{\mathbb{C}}\{W, f_z\}$  is a totally isotropic two-plane wherever it has rank two, PIC makes the second term strictly negative on that set. Thus such sections provide negative directions. One must establish negativity on a whole linear space, not merely exhibit unrelated negative vectors.

**Supplement (not independently green-marked).** The later argument in §5.2.3 constructs enough appropriate holomorphic sections using the splitting of bundles over  $S^2$  and dimension counting. It must also account for tangential symmetry directions, zeros, and the chart dependence of  $f_z$ . That construction is not reproduced here. The useful connection to remember is

$$\text{holomorphic isotropic sections} \quad + \quad \text{PIC} \quad \implies \quad \text{negative index directions.}$$

## 15 Formula sheet and reading checkpoints

### Core formulas

$$\begin{aligned}
E(u) &= \frac{1}{2} \int |du|^2, & \tau(u) &= \operatorname{tr}_g \nabla du, \\
\tau(u) = 0 &\iff \Delta_g u + \operatorname{tr}_g A(u)(du, du) = 0, & A &= -(D_X Y)^\perp, \\
\Theta_u(x, r) &= r^{2-m} \int_{B_r(x)} |du|^2, & \Theta'_u(x, r) &= 2r^{2-m} \int_{\partial B_r(x)} |\partial_r u|^2, \\
\Phi(u) &= 4(u_z, u_z) dz^2, & u \text{ harmonic} &\Rightarrow \bar{\partial}\Phi = 0, \\
A(u) &\leq E(u) \quad (\dim \Sigma = 2), & \Phi = 0 &\iff u \text{ weakly conformal}, \\
\mathcal{I} &= R_{1313} + R_{1414} + R_{2323} + R_{2424} - 2R_{1234}, & & \\
|R_{1234}| &\leq \frac{2}{3}(b-a), & b < 4a &\Rightarrow \text{PIC}.
\end{aligned}$$

### Distinctions to remember

1. **Critical versus minimizing:** first variation zero does not give a minimum; the second variation can have negative directions.
2. **Target versus domain variation:** weak harmonicity and domain stationarity are separate requirements at low regularity.
3. **Small energy versus small normalized energy:** the latter is the regularity quantity in dimensions above two.
4. **Weak versus strong compactness:** a weak limit may lose energy; minimizing-map compactness is strong locally.
5. **Zero measure versus lower dimension:**  $\mathcal{H}^{m-2} = 0$  alone does not give dimension at most  $m-3$ .
6. **Harmonic versus conformal:** on a general surface the Hopf differential can be nonzero; on  $S^2$  it must vanish.
7. **Bilinear versus Hermitian metric:** isotropy uses  $(z, z)$ , whereas positivity of curvature uses a Hermitian norm and pairing.
8. **Minimal versus minimizing:** a harmonic sphere is a branched minimal map but may have a large Morse index.